Inverted pendulum

Magdi Laoun, 17.12.2025

1 Abstract

An inverted pendulum is a pendulum that is held in equilibrium at the top in an unstable position. It can pivot at its base around an axis that is attached to a carriage capable of moving horizontally. With the help of a motor, the carriage moves to try to maintain this equilibrium position.

An angular sensor mounted on the carriage axis measures the pendulum's angle relative to the vertical in real time, and this data is processed by a microcontroller, which controls the motor to maintain balance.

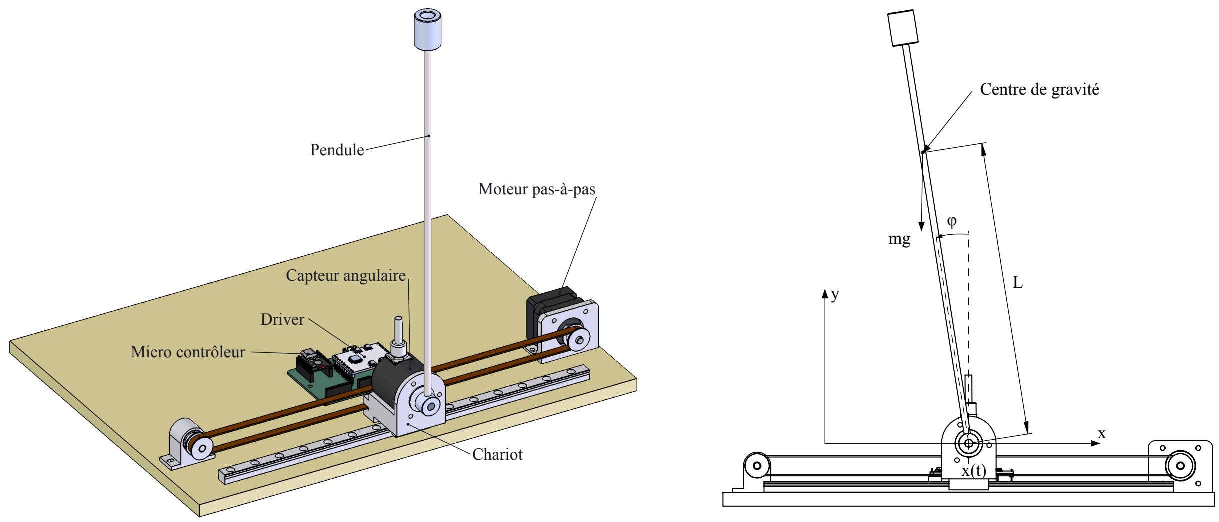

Figure 1: Geometry of the pendulum

2 Objectives

- Determine the movement of the carriage (and thus the motor) based on the pendulum's angle and position, so as to keep it vertical and centered on the rail.
- Determine the motor control coefficients needed to keep this pendulum in balance.

3 Data and hypotheses

- The motor has 200 steps per revolution and 16 microsteps per step, which amounts to 3,200 microsteps per revolution. The motor is a stepper motor. Consequently, the motor's position, speed, and acceleration do not depend on the load but solely on the electrical command provided by the controller.
- The motor pulley has a circumference of 40 mm. From this, we can deduce that the carriage moves 10 mm in 800 steps per revolution.
- The angle sensor mounted on the carriage, which measures the angular position of the pendulum, has a resolution of 10,000 counts per revolution.
- The total mass, moment of inertia, and center of gravity of the pendulum are known and provided by 3D modeling software.
- The pendulum's axis, at its base, is fixed by the carriage, with no play and no friction.
- The carriage moves only horizontally along the x-axis.
- All units used in the formulas and calculations are those of the International System of Units. For practical and readability reasons, different units may be used to denote a given value or result, such as millimeters for length, or grams

4 Settings and terminology

Paramètre	Signification
m	segment mass i
l	segment length i
p	Angular position of the pendulum
$\dot{\varphi}$	Angular velocity of the pendulum
$\ddot{\varphi}$	Angular acceleration of the pendulum
x	Carriage position
$\dot{x}$	Carriage velocity
$\ddot{x}$	Carriage acceleration

5 Kinematics

5.1 Lagrange's equations

We will solve the equations of motion using Lagrange's equations (energy balance).

$$\mathcal{L} = T - V$$

Where T is the kinetic energy of the system and V is its potential energy.

Since we are dealing with a stepper motor in which the rotor's acceleration is imposed by the motor and not caused by the load, we will assume that the mass of the carriage is zero. The kinetic energy of the system is therefore limited solely to the energy of the pendulum itself.

$$T = \frac{1}{2} m v_p^2$$

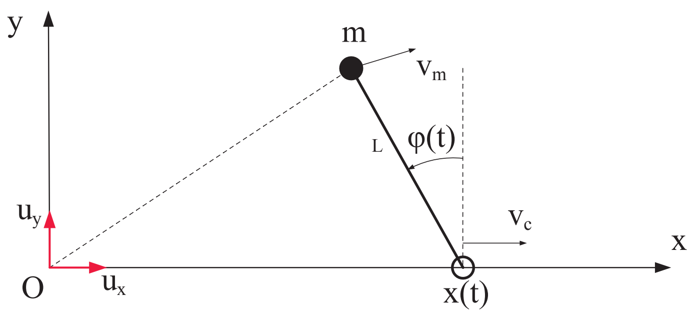

Figure 2: Geometry of the pendulum

Position of the pendulum

$x_p = x - l \sin \varphi$

$y_p = l \cos \varphi$

Velocity of the pendulum

$\dot{x}_p = \dot{x} - l\dot{\varphi} \cos \varphi$

$\dot{y}_p = -l\dot{\varphi} \sin \varphi$

$\rightarrow v_p^2 = \dot{x}_p^2 + \dot{y}_p^2 = \dot{x}^2 - 2\dot{x}l\dot{\varphi}\cos\varphi + l^2\dot{\varphi}^2\cos^2\varphi + l^2\dot{\varphi}^2\sin^2\varphi = \dot{x}^2 - 2\dot{x}l\dot{\varphi}\cos\varphi + l^2\dot{\varphi}^2$

Kinetic energy

Kinetic energy is therefore equal to: $T = \frac{1}{2} m\left(\dot{x}^2 - 2\dot{x}l\dot{\varphi}\cos\varphi + l^2\dot{\varphi}^2\right)$

Potential energy

As for potential energy, it is equal to: $V = mgl\cos\varphi$

Lagrangien

The Lagrangian is therefore written as: $\mathcal{L} = \frac{1}{2}m(\dot{x}^2 - 2\dot{x}l\dot{\varphi}\cos\varphi + l^2\dot{\varphi}^2) - mgl\cos\varphi$

5.2 Euler-Lagrange equations

Let's define the following vector:

$q = \begin{pmatrix} x \\ \varphi \end{pmatrix}$

The general form of the Euler-Lagrange equations is given by:

$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}}\right) - \frac{\partial\mathcal{L}}{\partial q} = u$

where u is the external generalized forces.

Since we are dealing with a stepper motor whose linear motion of the carriage is fully determined, we are not interested in the linear component along the x-axis; therefore, we will solve the Euler-Lagrange equation for the angular component only, and thus $u = 0$:

$\frac{\partial\mathcal{L}}{\partial\dot{\varphi}} = -ml\dot{x}\cos\varphi + ml\dot{\varphi}$

$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\varphi}}\right) = -ml\ddot{x}\cos\varphi + ml\dot{x}\dot{\varphi}\sin\varphi + ml^2\ddot{\varphi}$

$\frac{\partial\mathcal{L}}{\partial\varphi} = ml\dot{x}\dot{\varphi}\sin\varphi + mgl\sin\varphi$

The Euler-Lagrange equation is therefore written as:

$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}}\right) - \frac{\partial\mathcal{L}}{\partial q} = -ml\ddot{x}\cos\varphi + ml\dot{x}\dot{\varphi}\sin\varphi + ml^2\ddot{\varphi} - ml\dot{x}\dot{\varphi}\sin\varphi - mgl\sin\varphi = 0$

After simplifying, we get:

$$-\ddot{x}\cos\varphi + l\ddot{\varphi} - g\sin\varphi = 0$$

This gives us the following differential equation:

$$\ddot{\varphi} = \frac{g}{l}\sin\varphi + \frac{1}{l}\ddot{x}\cos\varphi$$

5.3 Linearization around the equilibrium point

Around the equilibrium point $\varphi \approx 0$, we can make the following approximations:

$\sin\varphi \approx \varphi$ et $\cos\varphi \approx 1$

This simplifies the equation above:

$$\ddot{\varphi} = \frac{g}{l}\varphi + \frac{1}{l}\ddot{x}$$

6 Matrix formulation

Let's define the following state variable: $X = \begin{pmatrix} \varphi \\ \dot{\varphi} \\ x \\ \dot{x} \end{pmatrix}$

The state variable is a variable that describes the pendulum's state at any given time t. The pendulum's state at time t is determined by the variables φ, $\dot{\varphi}$, x, and d

so that the equation of state can be expressed as follows:

$$\dot{X} = AX + B\ddot{x}$$

where A and B are matrices 4×4 to be determined. The equation can be written as follows:

$$\begin{pmatrix} \dot{\varphi} \\ \ddot{\varphi} \\ \dot{x} \\ \ddot{x} \end{pmatrix} = A \begin{pmatrix} \varphi \\ \dot{\varphi} \\ x \\ \dot{x} \end{pmatrix} + B\ddot{x}$$

where A and B are matrices. The equation of motion allows us to determine these matrices:

$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{g}{l} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 \\ \frac{1}{l} \\ 0 \\ 1 \end{pmatrix}$

This state equation is a linear quadratic differential equation, hence the name LQR, which stands for Linear Quadratic Regulation.

We will return to this equation of state later.

7 LQR Control

7.1 Introduction

Everything that has been done so far is preparatory work that will be used to achieve our goal here, namely, a control rule for the carriage acceleration based on the state variable that keeps the system in equilibrium.

$$\ddot{x} = -KX$$

where

$\ddot{x}$ is the acceleration that must be applied to the carriage to keep the pendulum in equilibrium

$X = \begin{pmatrix} \varphi \\ \dot{\varphi} \\ x \\ \dot{x} \end{pmatrix}$ is the system's state variable

$K = \begin{pmatrix} \text{kpa} & \text{kda} & \text{kpx} & \text{kdx} \end{pmatrix}$ are the control coefficients, and that is precisely what we are looking for.

The whole challenge lies in finding these four coefficients, which are key to successful control. We will examine this aspect in the next section.

7.2 Riccati function

Let:

ε_{p_1} and ε_{p_2}: the quadratic errors of the angles of the pendulum's two segments relative to its equilibrium position

The control parameter K is optimal if:

- These two quadratic errors are minimized over a certain period
- The motor's effort required to achieve this is minimized

The Ricatti function, named after its inventor, allows us to calculate this quadratic deviation as well as the motor's effort using an integral over a certain period. The control parameter K will be optimal when this result is minimized. The Ricatti function is as follows:

$$J = \int_0^\infty (X^T Q X + R\ddot{x}^2) dx$$

where:

- Q is a weighting matrix used to assign weights to any of the state variables φ, $\dot{\varphi}$, x, or $\dot{x}$
- R is a weighting coefficient used to assign a weight to the control cost

For given weights, the control is optimal if this function is minimized

The form of these two weights is as follows:

$$Q = \begin{pmatrix} q_\varphi & 0 & 0 & 0 \\ 0 & q_{\dot\varphi} & 0 & 0 \\ 0 & 0 & q_x & 0 \\ 0 & 0 & 0 & q_{\dot x} \end{pmatrix}$$

and

$$R = r$$

A good place to start is to ask:

- $q_\varphi = 1000$
- $q_{\dot\varphi} = 1000$
- $q_x = 10$
- $q_{\dot x} = 10$
- $r = 1$

7.3 Calculation of LQR control

From this, and according to Ricatti, there exists a matrix P 4×4 such that

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

Determining P by solving this matrix equation would, in theory, allow us to determine the coefficients K using the other Ricatti equation

$$K = R^{-1}B^T P$$

In practice, and in most cases, it is not possible to determine this matrix P analytically, given the nonlinear and differential nature of the relationships between the various parameters and variables.

However, it is possible to determine this matrix P numerically using computational tools such as MATLAB, which provides a specific tool for solving this problem, or by using a computer program that will be described below.

7.3.1 Differential equation, Riccati method

A Riccati differential equation has the following form:

$$\dot y = q_0(x) + q_1(x)y + q_2(x)y^2$$

where q_0, q_1 et q_2 are three known functions.

The Riccati differential equation also applies to matrices. In our case, we have

$$\dot P = A^T P + PA - PBR^{-1}B^T P + Q$$

with $\dot P \to 0$ when $t \to \infty$

To determine P, simply apply Euler's method, taking as the initial condition, for example, $P_0 = I$ (the identity matrix), and iteratively solve the following equation:

$$P_{k+1} = P_k + dt\dot P_k$$

Until $\left\|\dot P\right\| \to 0$ (matrix norm)

In practice, all you need to do is choose dt sufficiently small—for example, $dt = 0.001$—and you will see that the solution converges fairly quickly, after about 1000 iterations.

Note

Note that if the number of iterations is too high, rounding errors will cause the solution to diverge again.

7.4 Numerical example

Using code written in Swift for macOS, we calculated the various matrices A, B, and Q and performed the different operations needed to determine the matrix A and the coefficients K.

- length: $l = 300 \text{ mm}$
- gravity: $g = 9.81 \text{ m}/s^2$

These data (in SI units: kg, m, etc.) allow us to determine matrices A and B

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 32.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 \\ 3.33 \\ 0 \\ 0 \end{pmatrix}$$

Regarding the weighting matrix, we will prioritize angle and position over damping rate, using an effort cost of $r = 1$ as the baseline

$$Q = \begin{pmatrix} 1000 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1000 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Using a program code that allows us to manipulate matrices (addition, subtraction, multiplication, transposition, etc.) and a loop that calculates, iteratively and using Euler's method, the equation we saw in the 7.3.1 section, we obtain the following matrix P (rounded to the nearest integer):

$$P = \begin{pmatrix} 531 & 85 & -412 & -201 \\ 85 & 16 & -78 & -40 \\ -412 & -78 & 677 & 228 \\ -201 & -40 & 228 & 113 \end{pmatrix}$$

And finally, this allows us to determine the following LQR control coefficients:

$$K = \begin{pmatrix} 80.0 \\ 12.8 \\ -31.6 \\ -20.8 \end{pmatrix}$$

which allows us to determine the acceleration that must be applied to the motor to keep the pendulum in equilibrium (SI units: angle: rad, acceleration: $\frac{m}{s^2}$).

$$\boxed{\ddot{x} = -KX = -80.0 \cdot \varphi - 12.8 \cdot \dot{\varphi} + 31.6 \cdot x + 20.8 \cdot \dot{x}}$$

7.5 Practical test

With the values above, the device works perfectly. However, the device performs even better with the following coefficients, which were determined through trial and error:

$$\ddot{x} = -KX = -220.0 \cdot \varphi - 22.0 \cdot \dot{\varphi} + 224.0 \cdot x + 62.0 \cdot \dot{x}$$

This difference is likely due to the choice of values for the weighting matrix. In any case, the machine works perfectly whether you use the theoretical values or values determined by trial and error.

7.6 Program Convergence

We conducted several convergence tests on the Ricatti differential equation by varying the number of steps and the step size in the Euler method solution.

Nombre incréments	dt
50	0.04
100	0.02
1000	0.002

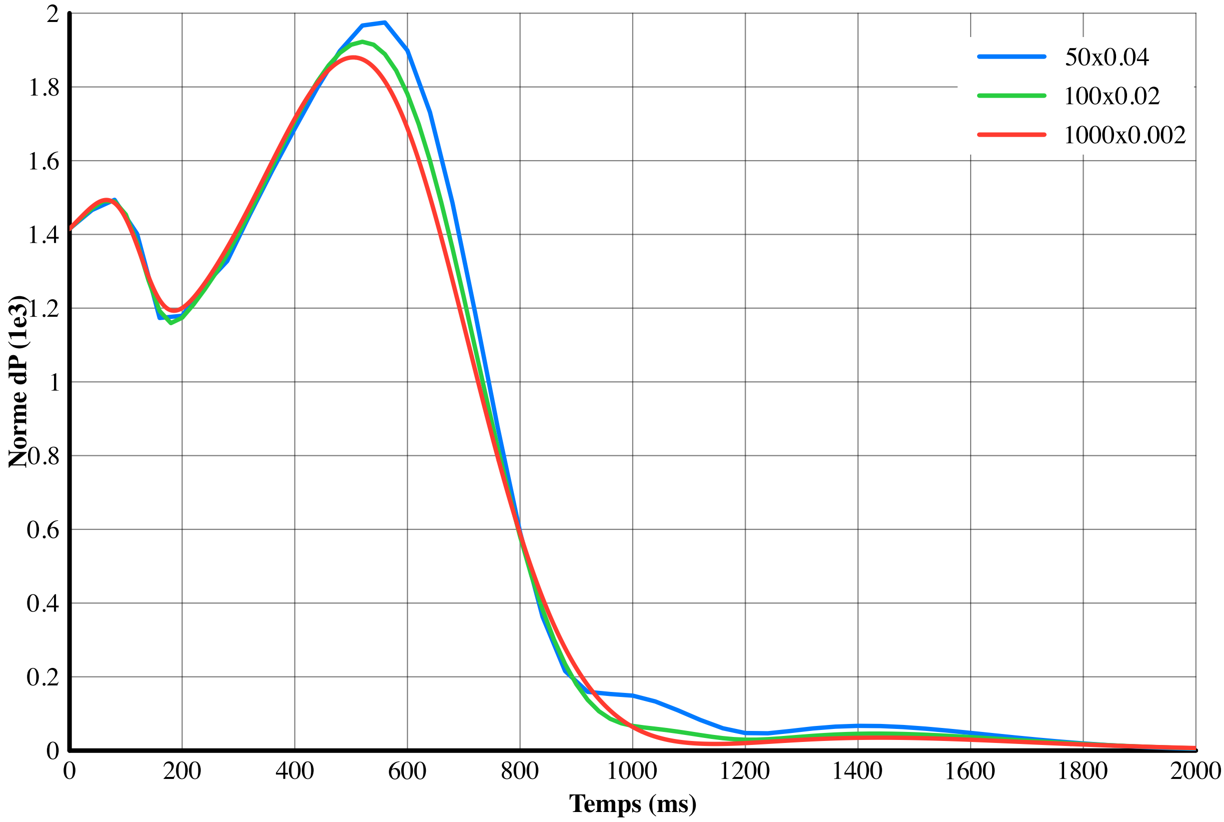

Figure 3: Norm of $\dot{P}$

What this graph shows is the norm of the matrix $\dot{P} = A^T P + PA - PBR^{-1}B^T P + Q$ during the iterations.

We can see that convergence is achieved relatively quickly, even with a large step size of $dt = 0.04$

8 Swing up

The "swing-up" is the process of raising the pendulum from its resting position at the bottom.

There are several methods for performing the swing-up.

A simple method involves moving the carriage back and forth, as one would on a swing, synchronized with the angle of the pendulum, so that with each movement of the carriage, the pendulum increases its speed enough to rise to the vertical position. There are two constraints:

- The carriage's movement must be synchronized with the pendulum's position: each time the pendulum passes through the bottom position, a movement is performed in the opposite direction of its velocity.
- When the carriage reaches a certain threshold angle (for example, 160°), it switches to top-position balance control mode. But be careful: the pendulum's speed must not be too high during the switch, otherwise balance cannot be maintained and the pendulum will continue its swing to the other side.

Another method involves controlling the carriage based on the balance of the pendulum's potential and kinetic energy. It is this method that we will examine below.

8.1 Energy balance method

The idea behind this method is to impart kinetic energy to the pendulum through the movement of the carriage; this energy will be converted into potential energy as the pendulum rises higher and higher, and the amount of energy imparted will depend on the position and velocity of the pendulum. Thus, the acceleration of the carriage is controlled by the pendulum's energy balance, so that the pendulum reaches the top as quickly as possible. Let E_p be the potential energy and E_c the kinetic energy of the pendulum

… To be continued….

9 Conclusions

From a kinematic perspective, determining the motion of the pendulum involves the Euler-Lagrange equations. This may seem complex at first glance, but it is ultimately manageable.

Furthermore, it is absolutely essential to clearly define the geometric data and, above all, the direction of the coordinates. Otherwise, a small sign error or a minor oversight can compromise the entire rest of the derivation.

The second part, which concerns the determination of the LQR control coefficients, is somewhat more complex. This derivation involves matrix operations as well as numerical computation to determine these control coefficients.

I would also like to highlight the achievements of the mathematician Riccati, who lived in the 17th century and did not have a computer to verify the accuracy of his formulas through practical testing.

Ultimately, we see that even for a problem as seemingly complex as the equilibrium of an inverted pendulum, a solution exists: we have been able to find a method to reliably determine the coefficients of the LQR control.